26900351 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMAT302 - AN INVITATION TO FUZZY MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. Give two examples of fuzzy variables in daily life. [K] / [CO1]
2. State the Third Decomposition Theorem for Fuzzy sets [U] / [CO1]
3. Give example of a fuzzy complement satisfying the skeleton property but not the involutive property. [A] / [CO2]
4. Write any two examples of Fuzzy t -norm. [U] / [CO2]
5. Define fuzzy relation. [K] / [CO3]
6. Define asymmetric binary relations for crisp and fuzzy cases. [K] / [CO3]
7. Define cycle in a fuzzy graph $G = (V, \sigma, \mu)$. [K] / [CO4]
8. Define a graph. [U] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Draw $\bar{A}(x)$ for the fuzzy set, $\mathrm{A(x)} = \begin{cases} 0 & \text{when } x \leq 20 \text{ or } x \geq 60 \\ \frac{x-20}{15} & \text{when } 20 < x < 35 \\ \frac{60-x}{15} & \text{when } 45 < x < 60 \\ 1 & \text{when } 35 \leq x \leq 45 \end{cases}$ [U] / [CO1]

10. Let $A, B \in \mathcal{F}(X)$. Then, for all $\alpha \in [0,1]$, show that $A \subseteq B$ if and only if ${}^{\alpha}A \subseteq {}^{\alpha}B$. [U] / [CO1]

11. a)Explain the concept of pseudo inverse of a decreasing generator. [A] / [CO2]

b)State characterization theorem of t - norms

12. Prove that max(a,b) and u_{max}(a,b) are the lowerbound and upperbound respectively of any valid t - conorm. [A] / [CO2]

13. Give a fuzzy membership matrix of a fuzzy compatibility relation and draw its graph. [A] / [CO3]

14. a) Explain how the composition can be performed using membership matrix. [A] / [CO3]

b) Compute $P \circ Q$ if $P = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.3 & 0 & 0.8 \\ 1 & 0 & 0.5 \end{bmatrix}$ and $Q = \begin{bmatrix} 0 & 0 & 0.5 \\ 0.6 & 0.2 & 0 \\ 0 & 0.7 & 0 \end{bmatrix}$

15. (a) Define fuzzy bridges and fuzzy cutvertices in a fuzzy graph G. [U] / [CO4]

(b) Give an example of a fuzzy block with a fuzzy bridge.

16. Define a fuzzy graph and explain its basic components with an example. [U] / [CO4]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Given a crisp function f : X → Y, write the formula that defines the extended function f(A) for a crisp set A ϵ P(X). What is the role of the supremum in the extension principle? How does the extension principle deal with two or more elements having the same image under a function? [U] / [CO1]

18. a) Let A and B be two fuzzy sets defined on a universal set X, with membership values for an element $x \in X$ given as $a = 0.6$ and $b = 0.7$. Calculate the membership values of the x in the standard union, algebraic sum, bounded sum and drastic union [U] / [CO2]

b) Let A and B be two fuzzy sets defined on a universal set X, with membership values for an element $x \in X$ given as $a = 0.4$ and $b = 0.9$. Calculate the membership values of the x in the standard union, algebraic sum, bounded sum and drastic union

19. a)Show that If a complement c has an equilibrium e_c, then $d_{e_c} = e_c$. [A] / [CO2]

b) Prove that for each $a \in [0,1], d_a = c(a)$ iff complement is involutive.

20. Let $X = \{x_1, x_2, x_3, x_4, x_5\}$ and let **R** be a fuzzy similarity relation on **X** defined by the matrix: $R = \begin{bmatrix} 1 & 0.8 & 0.4 & 0.5 & 0.8 \\ 0.8 & 1 & 0.4 & 0.5 & 0.8 \\ 0.4 & 0.4 & 1 & 0.4 & 0.4 \\ 0.5 & 0.5 & 0.4 & 1 & 0.5 \\ 0.8 & 0.8 & 0.4 & 0.5 & 1 \end{bmatrix}$ [A] / [CO3]

1. Find all similarity classes R_{x_i} for $i = 1, 2, 3, 4, 5$ as fuzzy sets on **X**.
2. Determine the level set Λ_R and write down the induced partition $\pi({}^{\alpha}R)$ for all threshold values $\alpha \in \Lambda_R$.

21. a)For a fuzzy graph $G = (V, \sigma, \mu)$ define the following . i) a partial fuzzy subgraph of G ii) fuzzy subgraph of G induced by $P \subseteq V$ [A] / [CO4]

b)For fuzzy graph $G = (V, \sigma, \mu)$, where $V = \{a, b, c\}, \sigma(a) = 0.5, \sigma(b) = 1, \sigma(c) = 0.8, \mu(ab) = 0.5, \mu(bc) = 0.8$ and $\mu(ac) = 0.1$, obtain fuzzy subgraph induced by following subsets of V. i) $\{a, b\}$ ii) $\{a, c\}$

c)Define the threshold graph of a fuzzy graph $G = (V, \sigma, \mu)$ corresponding to t, where $0 \leq t \leq 1$.

22. (a) Define a complete fuzzy graph. [U] / [CO4]

(b) Give an example of a complete fuzzy graph with a bridge.

(c) Give an example of a noncomplete fuzzy graph satisfying $\mu^{\infty} = \mu$.

[10x3 = 30]